

EvalúaYa — Algorithm technical specification

How the app decides Green, Yellow or Red. Every statement is taken from the working code (the source file is noted in grey).

How the final color is decided

The result is formed in two independent layers and the more severe one is kept. The safety rules (Layer 1) can only raise the level toward Red — they never make a result less severe than the AI suggested.

Order: read the earthquake data at the location (ShakeMap) → photos and answers go to the AI for a Green/Yellow/Red call → apply the deterministic rules → combine by taking the worse of the two.

source: *assessment.functions.ts* — `finalRisk = maxRisk(ai, rules)`

Layer 1 — Deterministic safety rules

Same inputs, same result, no AI. If any condition in the first group is true, the result is forced to Red. If any condition in the second group is true (and no Red rule fired), the result is forced to at least Yellow.

Condition → RED	Why
Structural system = unreinforced masonry (URM).	No steel reinforcement; sudden failure after strong shaking.
YES to ground-liquefaction signs.	Soil loses bearing capacity; settlement or tilting possible.
YES to pounding against a neighboring building.	Impact can cause severe, hidden structural damage.
YES to severe plumbing / gas damage.	Possible gas leak: an immediate life-safety hazard.
Strong shaking AND any reported structural damage.	MMI ≥ 8 or PGA ≥ 0.50g together with visible damage = critical combo.
Condition → ≥ YELLOW	Why
Moderate shaking: MMI ≥ 6 or PGA ≥ 0.25g.	Appreciable shaking; inspect more carefully even with no visible damage.
Severe shaking: MMI ≥ 8 or PGA ≥ 0.50g.	Higher caution floor based on the recorded intensity.
Spectral demand ≥ 0.40g at the building's period.	Buildings of that height felt the shaking especially hard.
Soft or very soft soil (low vs30).	Amplifies shaking and raises liquefaction / settlement risk.
More than 7 floors.	Greater consequence; upper floors need professional review.
Structural system CMF, CIW, PCF or RML.	Concrete / reinforced-masonry systems warrant extra caution.

If none apply, Layer 1 contributes Green and the AI's decision stands.

source: *safety-rules.ts* — `evaluateSafetyRules()`

USGS ShakeMap data (the data-driven layer)

If the resident shares their location, we read the official USGS ShakeMap grid for the active earthquake. The grid carries several co-registered layers, bilinearly interpolated to the building's exact point:

Layer	What it contributes
MMI (Mercalli intensity)	How strongly shaking was felt (scale I–XII).
PGA — peak acceleration (g)	Max instantaneous shaking; thresholds 0.25g and 0.50g.
PGV — peak velocity (cm/s)	Shaking energy; context for the AI.
SA(0.3 / 0.6 / 1.0 / 3.0 s)	Spectral acceleration by period: demand by height.
vs30 → soil class	Rock / stiff / soft / very soft (amplification, liquefaction).

Demand by height. We estimate the building's natural period as $T \approx 0.1 \times \text{number of floors}$ and pick the nearest spectral band ($\leq 0.45\text{s} \rightarrow \text{SA}0.3$; $\leq 0.8\text{s} \rightarrow \text{SA}0.6$; $\leq 2.0\text{s} \rightarrow \text{SA}1.0$; else $\rightarrow \text{SA}3.0$). That band's value is the shaking a building of that height actually experienced, and it feeds both the spectral-demand rule ($\geq 0.40\text{g}$) and the AI context.

Soil. $\text{vs}30 \geq 760 = \text{rock}$; $\geq 360 = \text{stiff}$; $\geq 180 = \text{soft}$; below = very soft. Soft soil raises the caution level because it amplifies shaking.

source: `shakemap.ts` — `seismicAt()`, `spectralDemand()`, `buildingPeriod()`, `soilClassFromVs30()`

Layer 2 — AI photo triage

Photos and answers go to a vision model (google/gemini-2.5-flash) acting as a structural engineer doing rapid post-earthquake triage (ATC-20 style), which picks exactly one level:

Level	Meaning
GREEN	No significant structural damage; appears safe to occupy.
YELLOW	Possible/moderate damage; restricted use, short essential tasks only.
RED	Serious damage or signs of collapse; unsafe, evacuate immediately.

The ground-motion context (MMI, PGA, PGV, period-matched spectral demand and soil class) is included in the prompt: strong shaking makes any reported damage weigh more, but shaking alone, with no observed damage, does not by itself force Red. The AI is conservative: when safety is uncertain, it does not choose green.

source: `assessment.functions.ts` — `SYSTEM_PROMPT` + `buildPrompt()`

How it combines — examples

AI says	Rule fired?	Final result
Green	URM → forces Red	RED
Yellow	Liquefaction = YES	RED
Green	PGA 0.55g + crack = YES	RED

Green	9 floors	YELLOW
Green	PGA 0.30g (no damage)	YELLOW
Yellow	No rule fired	YELLOW
Green	No rule fired	GREEN

source: *safety-rules.ts* — *maxRisk(a, b)* returns the more severe

Sources and limits

- ATC-20 (Applied Technology Council): rapid post-earthquake inspection method.
- USGS ShakeMap: official ground-motion grid for the active earthquake.
- Unreinforced masonry (URM) vulnerability: established seismic literature.
- AI vision model: google/gemini-2.5-flash via Lovable AI Gateway.

Limits: this is preliminary guidance, not a certification. It only evaluates the visible items you report; it does not inspect hidden elements or inside walls. It does not replace a licensed structural engineer or Civil Protection.